

Workflow: From an elevation model to a terrain map

terrain(v = , unit =)

calculate from the original elevation model, with elevations in meters and a known CRS; returns slope, aspect, TPI, TRI, roughness and flowdir; aspect is circular, so take `cos()` in radians rather than its mean

then

How do you want to show the surface?

on a flat map

shade(slope, aspect, angle, direction)

slope and aspect in radians; angle is the elevation of the light and direction its bearing, conventionally 315, because a light from the southeast inverts the relief

then

ggnewscale::new_scale_fill()

sits between two `geom_spatraster()` calls: the grey hillshade below, the colored layer above it with alpha, so the relief shows through the colors

in three dimensions

project() then rayshader::raster_to_matrix()

put horizontal spacing and elevation in compatible units, then let `raster_to_matrix()` handle orientation

then

sphere_shade() ray_shade() add_shadow()

then `plot_map()` for an image or `plot_3d()` for a scene; `zscale` is the horizontal cell spacing divided by the vertical unit, and `ray_shade()` is slow: build it small